CHAPTER 5

Generics and Habituals

We have argued above that generics express quantification, and the same is usually said about habituals. This quantifier is, as all agree, phonologically null. This means that it does not occur in the sentence; it needs to be derived somehow. What is the process by which this quantifier is produced?

We have in our arsenal two processes by which quantifiers are introduced: type-shifting and predicate transfer. These processes have different properties, as we have seen in chapter 2 above. In particular, they behave differently with respect to scope: quantifiers introduced by type-shifting receive narrow scope only, whereas those introduced by predicate transfer engage in free scope ambiguities in transparent contexts.

The first step in an account of the origins of the generic and habitual quantifiers will, therefore, be to look at their scopal properties. We will consider generics first, and then move on to habituals.

1 Generics and Scope

What is the scopal behavior of generics? Do they display normal scope ambiguities, no scope ambiguities, or are they somewhere in between? This question is surprisingly hard to answer. It is not immediately clear whether generics can interact scopally even with a simple operator, such as negation.

1.1 *Negation*

Carlson (1977) argues that generics do not exhibit any scope ambiguities. As an example, he considers the effect of negation. Carlson notes that (230a) is ambiguous between readings that can be paraphrased as (230b) and (230c).

(230) a. Bill doesn't like all wombats
b. For all *x*, if *x* is a wombat then it is false that Bill likes *x*.
c. It is false that, for all *x*, if *x* is a wombat then Bill likes *x*.

In contrast, (231) is not ambiguous; it can only get a reading that can be approximated by (230b), not the reading approximated by (230c).[1]

1 I say "approximated", since, of course, the quantifier in (231) is the generic quantifier, not the universal one, and it allows exceptions.

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(231) Bill doesn't like wombats.

Hence, Carlson concludes that generics are not ambiguous with respect to negation.

This is not the end of the story, however. Krifka et al. (1995) reach the opposite conclusion, and argue that generics *are* ambiguous with respect to negation. They discuss (232), which, they claim "can mean either that cows do not have the habit of eating nettles or that they have the habit of not eating nettles" (p. 123).

(232) Cows do not eat nettles.

Krifka et al. do not explain exactly what the truth-conditional difference between these two readings is (we will return to this issue in a moment), but it is fairly clear that there is, indeed, a real ambiguity here.

Hence, there is evidence that generics are *not* ambiguous with respect to negation, and there is evidence that they *are*. How can we resolve this conflict?

One possible solution might be to deny that (231) has any quantificational reading. We could say that (231) simply expresses predication of the kind *wombat*, expressing the claim that it does not have the property of being liked by Bill. According to this interpretation, (231) is not about individual wombats at all. Bill does not even have to know any wombats for the sentence to be true; he just has to have a general dislike toward the species. Since there is no quantifier involved, there is no puzzle why there is no scope ambiguity. In contrast, when a quantifier does show up, as in (232), the ambiguity is there.

It cannot be denied that (231) does, indeed, have such a reading. Moreover, this is probably the preferred interpretation. So, perhaps it really is the only reading, and (231) does not involve quantification.

But this proposed solution would only push the problem one step further: why does (231) have only a kind reading? We know that *like* can, in fact, take individual objects:

(233) Bill likes $\left\{\begin{array}{l}\text{Mary}\\ \text{every politician}\\ \text{most colleagues}\end{array}\right\}$.

There is no way to interpret (233) as kind predication, but only as a statement about Bill's feelings toward individuals.

Moreover, a quantificational reading of (231) *is* possible, if somewhat unnatural, if we add an overt adverb of quantification, as in:

(234) John usually likes wombats (but he hates my pet wombat).

Many more examples are provided by Diesing (1992, 113–114):[2]

(235) a. I usually like pictures of manatees.
b. I usually love sonatas by Dittersdorf.
c. I usually appreciate good jokes about violists.
d. I generally hate articles about carpenter ants.
e. I generally detest operas by Wagner.
f. I usually dislike movies about vampires.
g. I generally abhor books about Brussels sprouts.
h. I usually despise paintings of Chester A. Arthur.
i. I generally loathe stories about stockbrokers.

Incidentally, there are two options to negate (234) (and the sentences in (235)), depending on whether negation is placed before or after the quantificational adverb (in surface sturcture), resulting in two different readings:

(236) a. John doesn't usually like wombats.
b. John usually doesn't like wombats.

Sentence (236a), but not (236b), is compatible with a situation in which John is indifferent towards wombats.

Even if it turns out that (231) does not, in fact, have a quantificational reading, there are a great many other cases with a generically interpreted object, where it is clear that we do have a characterizing reading:

(237) a. Neil doesn't fly Apollo missions.
b. This security guard doesn't identify employees.
c. This program doesn't support women.
d. This court doesn't hear appeals.
e. The big gun doesn't mark Royal events.

Note that these cases allow exceptions: Neil may fly an Apollo mission on a rare occasion, the security guard may identify employees from time to time, etc. The question, then, is this: do we have a scope ambiguity here?

2 Diesing's examples actually involve indefinite singular objects, rather than BPs, but the judgments are the same.

This question is not so easy to answer as it stands. But note what happens if we stress the negation:

(238) a. Neil $\left\{\begin{matrix}\text{DOESn't}\\ \text{does NOT}\end{matrix}\right\}$ fly Apollo missions.

b. This security guard $\left\{\begin{matrix}\text{DOESn't}\\ \text{does NOT}\end{matrix}\right\}$ identify employees.

c. This program $\left\{\begin{matrix}\text{DOESn't}\\ \text{does NOT}\end{matrix}\right\}$ support women.

d. This court $\left\{\begin{matrix}\text{DOESn't}\\ \text{does NOT}\end{matrix}\right\}$ hear appeals.

e. The big gun $\left\{\begin{matrix}\text{DOESn't}\\ \text{does NOT}\end{matrix}\right\}$ mark Royal events.

Now we get a much stronger reading, which allows no exceptions: Neil doesn't fly *any* Apollo missions, the security guard doesn't recognize *any* employees, etc. I think the same effect is also present in (231), but it is hard to detect, because of the dominance of the kind reading of this sentence. It is, however, clearly present in (232):

(239) Cows do NOT eat nettles.

While (232) tolerates exceptions, the truth of (239) permits no cow to eat nettles.

What is the nature of this effect of focus on the interpretation of negated generics? Specifically, and most relevantly for our purposes, is it a scope ambiguity? Following Cohen (2001a), I suggest an answer to the first question that will entail an affirmative answer to the second one.

Consider (237b), for example. Let us first assume that negation takes narrow scope. In this case, we get the reading where, in general, if x is an employee, the security guard doesn't identify x. Of course, the domain of quantification is restricted only to relevant employees, say employees that tried to pass through the door guarded by the security guard.

It is well established that focus can further restrict the domain of quantification. For example:

(240) a. This security guard doesn't identify the [signatures]$_F$ of employees.
b. This security guard doesn't [identify]$_F$ the signatures of employees.

Here, the role of focus is to introduce alternatives; the disjunction of the alternatives restricts the domain of quantification (Rooth 1985). Thus, we interpret the sentence as follows: in general, if x is a security guard, and x satisfies the disjunction of the alternatives, then the security guard doesn't identify x.

In (240a) we consider alternative means of identification; and the sentence says that employees who are identified by the security guard are not identified on the basis of their signatures. In contrast, (240b) is about employees who present their signatures for inspection, and says that, in general, they are not identified.

Let us now consider what happens when we focus the negation itself:

(241) The security guard [doesn't]$_F$ identify employees.

In this case, I suggest that negation takes wide scope. Note that there is focus on the auxiliary, and therefore no focus inside the scope of the generic. Hence, no alternatives are introduced, and the set of alternatives is simply the set consisting of a single property: the property of not being identified by the security guard.[3]

What happens when the set of alternatives is a singleton? Sentence (241) is then interpreted as follows (with wide scope negation): it is not the case that, in general, if x is an employee and the security guard identifies x, then the security guard identifies x. This will of course be trivially false, so long as the security guard identifies at least one employee. Hence, we get the interpretation that the truth of (241) requires the security guard not to identify *any* employee.

For another example, consider the following exchange:

(242) A: I just saw a documentary about the platypus. What a funny mammal—it lays eggs!
B: Don't be silly. Mammals [don't]$_F$ lay eggs.

B's response means that it is not the case that there exists x s.t. x is a mammal and x lays eggs. Is it therefore, of course, false. Crucially, B cannot be understood as making the (true) statement that it is not the case that, in general, if x is a mammal then x lays eggs.

This, then, is the nature of the ambiguity of (232) above. Under one reading, negation takes narrow scope, and the sentence says that, in general, cows do

3 Compare Krifka (1999, p. 269): "the alternatives that are introduced by focus are used by … focus-sensitive operators … No alternatives are projected beyond that".

not eat nettles. But when negation is focused and takes wide scope, the sentence expresses the negation of the existential statement that some cows eat nettles, i.e. the claim that *no* cows eat nettles.

Hence, contrary to Carlson's claim, we can conclude that generics do interact scopally with negation.

1.2 *Opaque Contexts*

Carlson (1977) notes that (243), which employs an overt quantifier, is ambiguous.

(243) John believes that all professors are insane.

Under one reading, the quantifier has narrow scope with respect to the belief operator, and the sentence means that John has the following belief: for all x, if x is a professor then x is insane. Under the other reading, the quantifier has wide scope, and the sentence means that for all x, if x is a professor, then John has the following belief: x is insane.

In contrast, the generic (244) only has the first type of reading.

(244) John believes that professors are insane.

Sentence (244) can only mean that John believes the following: in general, if x is a professor then x is insane. It cannot mean that, in general, if x is a professor, then John believes that x is insane.

Judgments are rather hard, as Carlson himself admits. Let us try to find an example with clearer judgments.

(245) The King believes that every enemy spy is loyal to him.

Under one reading, the King has a rather irrational belief: "for all x, if x is an enemy spy, then x is loyal to me." Under the other, more plausible reading, the King is rational, but has simply failed to expose the spies in his camp. Thus, for every x, if x is an enemy spy, the King has the following belief: "x is loyal to me."

If we change the quantifier from the determiner *every* to the adverb *always*, we can see the two readings in surface structure, depending on whether the adverbial is inside or outside the intensional context:

(246) a. The King believes that an enemy spy is always loyal to him.
b. The King always believes that an enemy spy is loyal to him.

Sentence (246a) describes an irrational king, whereas (246b) is perfectly compatible with a rational king (who, presumably, has an incompetent counter-intelligence service).

What about the generic?

(247) The King believes that enemy spies are loyal to him.

Indeed, this only has the irrational reading. It can only mean that the King does not understand what it means to be a spy, not that he simply failed to expose the spies.

Hence, Carlson is right: generics do not scope out of opaque contexts.

1.3 *Transparent Contexts*

In contrast, in transparent contexts, generics exhibit the usual scope ambiguities (Schubert and Pelletier 1987):

(248) a. Canadian academics are supported by a single granting agency.
b. Storks have a favorite nesting area.
c. Dogs have a tail.
d. Sheep are black or white.
e. Whales are mammals or fish.

Of course, some readings are implausible (e.g., the reading of (248c) where the indefinite takes wide scope) but they are all, in principle, available.

The conclusion of this section is that generics do exhibit scope ambiguities, but they are more restricted than these of overt quantifiers; specifically, generics cannot scope out of opaque contexts.

2 Habituals and Scope

It is commonplace to treat generics and habituals on a par, and many authors do not bother to distinguish between them. Intuitively, the two phenomena do have a great deal in common: both involve no overt quantifier, and both seem to imply some sort of lawlike, permanent regularities, yet both tolerate exceptions.

There are, however, important differences between the two constructions. Of particular relevance to our discussion are the different extents to which generics and habituals exhibit scope ambiguities. Krifka et al. (1995) note that the habitual quantifier takes narrow scope with respect to an indefinite. Thus,

(249) is odd, because its only interpretation is that there is a cigarette that John habitually smokes.

(249) #John smokes a cigarette.

Since normally cigarettes are smoked only once, the sentence is pragmatically odd. Crucially, (249) cannot get the pragmatically plausible reading, where John habitually smokes a cigarette (different cigarettes on different occasions).

It should be pointed out that some informants do manage to get this reading, but I believe that they do this by accommodating an appropriate context (see below), in order to avoid the oddity. Indeed, (250) is not odd, hence does not get accommodated. And all informants agree the only reading it can get is that there is one brand of French cigarette that John habitually smokes, i.e. the habitual takes narrow scope.

(250) John smokes a brand of French cigarette.

Krifka et al. make a similar point. They note that, in contrast with cigarettes, pipes *are* smoked multiple times, and consequently (251) is fine.

(251) John smokes a pipe.

Again, the habitual takes narrow scope: (251) means that there is a pipe that John habitually smokes,[4] but this is no longer a pragmatic anomaly.

In this, the habitual differs from an overt adverbial quantifier. For example, the sentences in (252) *are* ambiguous.

(252) a. John never smokes a cigarette.
b. John smokes a cigarette every day.

Sentence (252a) can have either the implausible reading where there is one cigarette that John never smokes, or the much more reasonable reading where there are no occasions of John smoking. Similarly, (252b) can mean either that there is (implausibly) one cigarette that is smoked every day, or (much more plausibly) that on every day there is a cigarette that John smokes.

4 Of course, this does not exclude the possibility that there are additional pipes that John habitually smokes.

A similar phenomenon is observed when the habitual is restricted overtly, or even by the context. In such cases, habituals *are* ambiguous between wide and narrow scope readings.

Consider:

(253) a. John smokes a cigarette when he is nervous.
b. Q: What does John do when he is nervous?
A: He smokes a cigarette.
c. John bakes a cake for dinner (Hans Kamp, pc).

In (253a), a *when*-clause introduces the restrictor of the quantifier. This, apparently, is sufficient to allow the habitual to receive wide as well as narrow scope, and the sentence is ambiguous. Wide scope may even be enabled by contextual restriction. In (253b), the question sets up a restriction of the habitual quantifier, so that it ranges over situations in which John is nervous. Consequently, the answer is ambiguous, and can easily receive the plausible wide scope reading. Similarly, (253c) provides a restriction of the quantifier over dinner situations, and, consequently, a wide scope interpretation is easily available.

Things are even more complicated, in fact. As we have seen in chapter 3, section 2.2 above, if the indefinite is a bare plural, the habitual can have only wide scope, and it is the narrow scope reading that is unavailable. Thus, (254) means that John is habitually in a situation where there are some cigarettes that he smokes. It does not, indeed *cannot* get the interpretation corresponding to (249), namely that there are some cigarettes that John habitually smokes.

(254) John smokes cigarettes.

3 Types of Explanation

3.1 *The Facts so far*

Let us recap the scope facts we have observed so far. Generics exhibit scope ambiguities, with one significant limitation: they cannot scope out of opaque contexts. In contrast, habituals receive narrow scope only, except when they are overtly or contextually restricted, in which case they may take wide scope, and with bare plurals, where they *have* to take wide scope.

Surprisingly few accounts of these puzzling differences in scopal behavior between generics and habituals have been offered in the literature. Let us take a look at two of them.

3.2 *Number*

Krifka et al. (1995) assume that generics have normal scope ambiguities, and focus on the difference between sentences such as (249) and (254). They propose to account for the difference on the basis of the number of the indefinite: singulars allow only narrow scope to the habitual, whereas plurals allow both readings.

Krifka et al. assume that unrestricted habituals express quantification over a situation variable; this variable may be a simple situation, but it may also be a *sum* situation, i.e. a situation composed of a plurality of simple situations.

There are two ways in which quantification over sum situations can be acceptable. One way is to associate the sum situation with a sum individual, as in (254). In this case, every simple part of the sum situation is associated with a simple part of the sum individual. This is how we get the interpretation where different cigarettes are smoked on different occasions.

An alternative is to associate one simple individual with the sum situation as a whole, as in (251). This is how we get the narrow scope interpretation of the habitual, namely that a single pipe is associated with multiple smoking events.

However, quantification over sum individuals is not compatible with associating a different simple individual with each one of the simple situations in the sum. This is the interpretation that would make sense of (249), and the fact that it is unavailable explains the unacceptability of the sentence. Krifka et al. assume that explicitly restricted habituals, as in (253a), can only quantify over simple situations. Presumably, the same would hold for overt adverbial quantifiers, as in (252), and cases where the restriction is provided by the context, as in (253b) and (253c).

There are, however, problems with Krifka et al.'s account. First, the difference between unrestricted habituals on the one hand, and restricted habituals (and overt Q-adverbs and certain contexts) on the other hand, is unmotivated. Why is is that the former, but not the latter, may quantify over sum situations?

Second, while Krifka et al. provide an explanation for why the habitual in (254) *may* have wide scope, they fail to explain why it *must*. Since habituals, by this account, may quantify over simple situations as well as sum situations, why can't the bare plural scope over the habitual, associating each of the simple individuals in its sum with a simple situation?

Third, consider the following sentence:

(255) #John smokes some cigarettes.

The direct object is plural, just like the direct object of (254). If, as Krifka et al. 1995 claim, all that matters were the number of the TNP, it should behave scopally just like it does in (254), namely take narrow scope with respect to the habitual. But the facts are exactly the reverse: (255) can only have the implausible reading that there are some cigarettes that John habitually smokes,[5] and does not have the reading where John is habitually in a situation where there are some cigarettes that he smokes.

3.3 *A Verb-Level Null Operator*

A different account has been proposed by Van Geenhoven (2004). She does not deal with generics, but only habituals, and assumes a phonologically null operator, whose role is to express habituality.[6] This operator applies directly at the level of the verb—hence, any operator that scope over the verb, also scopes over this operator. This results in habituals having narrow scope only.

This account is, however, problematic, on both theoretical and empirical grounds. From a theoretical point of view, this proposal is not well motivated. Why would the habituality operator apply at the verb level, and not scope freely like most other operators? Moreover, the explanation has empirical difficulties: Van Geenhoven's proposal fails to account for the examples in (253), where a habitual does receive wide scope.[7]

3.4 *A Syntactic Account?*

Perhaps the explanation of these facts is syntactic, rather than semantic, in nature. No such idea has been explored, as far as I am aware, but suppose that implicit quantification (generic or habitual) applies to a phrase structure tree at the lowest level possible. Habituals quantify over the event argument, which is inside the VP. Hence, these operators apply at the level of the VP, so that everything that has scope over the VP also scopes over them—this is why habituals receive narrow scope only. In contrast, generics quantify over the subject. Hence, they apply at the IP level, and are therefore scopally ambiguous. The relevant fact about opaque contexts is not their opacity, but the fact that they are embedded; and the generic cannot scope over the embedding verb.

Would such an account be syntactically well motivated? I am not sure: many questions would remain to be worked out. For example, why do implicit oper-

5 Under the taxonomic interpretation, where *some cigarettes* is interpreted as brands of cigarette, the sentence actually makes perfect sense.

6 In fact for Van Geenhoven habituality is reduced to iterativity; we will return to iterativity and to Van Geenhoven's theory in chapter 6.

7 Nickel (2016) also proposes that the generic operator applies at the level of the predicate, but not out of considerations of scope: he argues that **gen** is a distributivity operator. I will address his account in chapter 7, section 1.2.

ators have to operate at the lowest level possible, while explicit operators are not so restricted? What, precisely, is the syntactic location of these operators, and why can't they QR?[8]

However, we needn't worry about the theoretical justification for such a theory, because it is empirically incorrect. The crucial test cases are sentences where the object is interpreted generically, and takes wide scope. This should not be possible according to the syntactic account, since the object is inside the VP, hence the generic operator should also apply at this level, and receive narrow scope only. However, it is easy to construct this type of sentence:

(256) a. $\left\{\begin{array}{l}\text{A well trained team flew}\\ \text{Three astronauts flew}\end{array}\right\}$ Apollo missions.
b. A personal code identifies admissible users.
c. Two bodyguards protect Ministers, but a single bodyguard protects Deputy Ministers.

Under their most plausible readings, sentence (256a) means that, in general, for each Apollo mission there was a team that flew it; (256b) refers to a different code for different users; and in (256c), different bodyguards are assigned to different politicians. Hence, in all these sentences the object is interpreted generically, and receives wide scope.

The sentences above are manufactured examples. But one can also find naturally occurring examples on the Web:

(257) a. Family planning helps women or girls to avoid pregnancies and continue with their education.[9]
b. Three judges sit together to hear appeals from decisions of the Trial Division.[10]
c. At sexual maturity a modified anal fin distinguishes the male Guppy.[11]
d. Twenty one gun salutes were fired from the large guns to mark Royal events.[12]

8 Rimell (2004) assumes that the habituality quantifier applies at the level of AspP; she further assumes that no element can scope below AspP. She does not, however, provide an argument for either of these claims.

9 https://repository.unam.edu.na/bitstream/handle/11070/421/nandjila2008.pdf?sequence=1&isAllowed=y

10 http://www.isn.net/cliapei/pub/78b.htm

11 https://www.practicalfishkeeping.co.uk/features/the-guppy-first-loves-never-die/

12 http://www.caithness.org/caithnessfieldclub/bulletins/1996/artillery_batteries_castletown_and_mey.htm

The dominant reading of (257a) is that, if x is a woman or a girl, family planning helps x, not that family planning helps women, or family planning helps girls; (257b) allows different judges for different appeals; (257c) is about different fins for different guppies; and (257d) is about different gun salutes for different occasions of royal events.

We can conclude, then, that all three suggestions made above do not seem to explain the scopal facts about generics and habituals successfully—can we do better?

4 Reinterpretation Mechanisms Revisited

4.1 *The Introduction of Implicit Quantifiers*

Explicit quantifiers, as in (252), are produced by the speaker, and appear in the input to the hearer. And, as we have seen, they exhibit the normal scope ambiguities. In contrast, generics and habituals involve implicit quantification: the quantifier is phonologically null. But this means that, in a very real sense, the quantifier is not "there"; it is not produced by the speaker, nor does it appear in the input to the hearer. How, then does it appear in the interpretation? Where does it come from?

Sometimes, although the quantifier itself is implicit, its presence is indicated by explicit means, such as the occurrence of an overt restrictor, as in (253a). Even context, as in (253b) and (253c), may indicate the presence of the quantifier. In such cases, the quantifier is, in a sense, really "there", as if it were pronounced overtly. Hence, again, scope ambiguities are normal.

However, when there are no such overt means, the presence of the quantifier involves a change of meaning from the original input, where there is no quantifier, to the semantic representation, where a quantifier does occur.

However, in many accounts of generics and habituals this simple fact is ignored. The implicit assumption appears to be that, in some sense, the quantifier is already "there", and no special mechanisms are needed to introduce it. In other words, the only difference between overt quantifiers and the generic/habitual quantifier is phonological: the former have phonological content, while the latter are "null"—we can't hear them. But what, exactly, does it mean for a quantifier to be phonologically null?[13]

13 As mentioned above, some researchers (Leslie 2007b; Sterken 2015a) attempt to derive the meaning of the generic quantifier from the fact that it is phonologically null; however, they still assume that the quantifier is essentially "there", and do not discus the mechanisms by which it is derived. A notable exception is Nickel (2016); I will discuss his theory in chapter 7, section 1.2 below.

4.2 *Null Categories in Syntax*

The idea seems to come from syntax, where there is a long history of theories assuming empty categories of various kinds: traces (or unpronounced copies), PRO, pro, etc. Syntacticians propose that semantic interpretation makes use of these traces, so they must be available to semantics. How is this possible? The common answer is that such empty categories are simply "there": they have no phonological content, but are syntactically and semantically significant.

Consider, for example, the remarkable (if controversial) phenomenon of *wanna* contraction. In colloquial English, (258a) can quite acceptably be paraphrased as (258b).

(258) a. I want to come to the party.
b. I wanna come to the party.

Strikingly, however, (259b) is not an acceptable paraphrase of (259a), and is, in fact, ungrammatical.

(259) a. Who do you want to come to the party?
b. *Who do you wanna come to the party?

The common argument is that *want to* cannot be turned into *wanna* if there is an empty element between *want* and *to*; and in (259a) there is such an unpronounced element, a trace (or an unpronounced copy), so its surface structure is actually (260).

(260) Who do you want *t* to come to the party?

Such examples have been taken to indicate that traces are very real: you can't hear them, but they are there all the same.[14] One almost feels that, like the Emperor's new clothes, if you can't see them, you must be a fool. Yet we mustn't lose sight of the fact that the Emperor is, in fact, naked: when we hear a sentence, what we are presented with is an expression without empty categories.

In fact, Chomsky (1995) argues that, at PF, empty categories are deleted, in accordance with his proposed Full Interpretation principle: "if a symbol in a representation has no sensorimotor interpretations, the representation does not qualify as a PF representation" (p. 27). Since what we hear is fed by PF,

14 This account of *wanna* contraction is, in fact, quite controversial, but I will not join the debate here.

and empty categories are deleted at PF, it follows that empty categories are not present in the input we hear, not even in some obscure “abstract” sense. What is often not appreciated is the fact that if hearers do not hear these elements in the input, they need to reconstruct them somehow.[15]

One might think that it does not really matter how we decide to treat empty categories. We may decide to treat empty categories as really “there”, though inaudible; alternatively, we may choose to accept that they are not “there”, and are reconstructed by the hearer. But is there an empirical fact that can distinguish between the two views?

Such an empirical fact would be provided by syntactic phenomena that are only sensitive to material that is really “there”. If there are syntactic, non-phonological rules that ignore unpronounced material and only apply to material that can actually be heard, such rules would provide evidence that the empty categories are not really “there”, not only phonologically but also syntactically.

It appears that there are, in fact, such rules. Holmberg (1999) notes that in Scandinavian languages, object shift is impossible if there is material inside the VP preceding the object. For example, in Swedish, the object cannot shift across an indirect object, from its position in (261a) to its position in (261b).

(261) a. *Jag gav inte Elsa den.*
I gave not Elsa it

b. **Jag gav* ***den**$_i$.*
I gave it

Similarly, while (262a) is fine, (262b), where the object has shifted across a verb particle, is bad.

(262) a. *Dom kastade inte ut mej.*
they threw not out me

b. **Dom kastade* ***mej**$_i$ *inte ut* ***t**$_i$.*
they threw me not out

15 Syntacticians who are interested in ellipsis often do, in fact, discuss the need to reconstruct the elided material (see, for example, Wasow 1972; Williams 1977; Haïk 1987; Kitagawa 1991; Fiengo and May 1994).

However, if the intervening material is phonologically null, object shift is perfectly acceptable; thus, the following sentences are good:

(263) a. *Vem*$_j$ *gav du* ***den***$_i$ *inte* t_j $\boldsymbol{t_i}$*?*
who gave you it not
'Who didn't you give it to?'

b. *UT*$_j$ *kastade dom* ***mej***$_i$ *inte* t_j $\boldsymbol{t_i}$ (*bara ned för trappan*).
out threw they me not only down the stairs
'They didn't throw me OUT (only down the stairs).'

Thus, Holmberg's Generalization appears to provide evidence that empty categories are not really "there"; they need to be reconstructed by the hearer.[16]

Reconstructing empty categories is not a trivial task: indeed, according to the Trace Deletion Hypothesis (Grodzinsky 1986, 1995), agrammatic aphasics are unable to do it.[17] So why do it at all? Why is it that, when we hear (264a), we reconstruct the phonologically null element PRO as the subject of *leave*, and derive (264b)?

(264) a. John wants to leave.
b. John$_i$ wants PRO$_i$ to leave.

For another example, consider (265a): why do we reconstruct a phonologically null trace as the object of *love*, and generate (265b) as the result?

(265) a. Who does Mary love?
b. Who$_i$ does Mary love t_i?

The answer is that, otherwise, syntactic constraints would be violated, and the sentences would be ill-formed. Thus, (264a) is ill-formed, because the inner clause lacks a subject, hence PRO is introduced to "save" the situation. Similarly, (265a) is bad, because the transitive verb lacks an object; hence, the trace is introduced, and the resulting structure, (265b), is fine.

16 There are alternatives to Holmberg's theory (e.g., Bobaljik 2002; Erteschik-Shir 2005; Fox and Pesetsky 2005). His empirical generalization, however, appears to be largely accepted.

17 Ironically, the name of the theory implies that traces are "there", and agrammatic aphasics somehow delete them. Of course, this is not the actual content of the theory: rather, agrammatics cannot recover traces that are not present in the sentences they hear.

It should be emphasized that, of course, introducing phonologically null elements is only one of several possible ways to save the situation. There are competing syntactic theories in which no such elements are posited. But the point of these examples is not to argue for a particular syntactic theory, but rather to show the rationale behind the postulation of empty categories: if they exist, the reason for their introduction is to save a sentence that, otherwise, would be bad.

4.3 *Null Categories in Semantics*

There are a few cases where null categories have been proposed on semantic, rather than syntactic grounds. But the reason for their introduction is similar: to save a sentence that, otherwise, would be unacceptable.

A classic case is that of quantificational domain restriction. It is widely agreed that when a quantifier is used in natural language, its domain is assumed to be restricted by the context. More specifically, the context provides a set, which is not included in the surface structure of the sentence, and the domain of the quantifier is understood to be intersected with this set.

For example, Neale (1990) writes:

> Suppose I had a dinner party last night. In response to a question as to how it went, I say to you:
>
> (266) (Neale's example #2) Everyone was sick.
>
> Clearly I do not mean to be asserting that everyone in existence was sick, just that everyone at the dinner party I had last night was.
>
> pp. 95–96

Sentence (266) does not provide any explicit restirction on the domain of the quantifier. Why, then, is the sentence not understood to express a statement about everyone in existence? Presumably, because such a statement would be blatantly false—it's a statement that anyone would immediately recognize as being false, and nobody in their right mind would have any reason to utter. In Gricean terms, such a statement would not merely violate the maxim of Quality—it would flout it. Therefore, we look to the context for a domain, which provides us, in this case, with the set of people at the dinner party, and that serves as the domain of the quantifier.

Similarly, Recanati (1996) notes that "when we say 'The burglar took everything', we have the feeling that 'everything' ranges over the domain of valuable objects in the house—not everything in the world" (p. 445). Again, saying that the burglar took everything in the world would be a nonsensical statement, hence the context provides us with an acceptable domain for the quantifier to range over.

Contextual restriction operates even when the sentence itself does provide some restriction. For example, Lewis (1996) writes: "If I say that every glass is empty, so it's time for another round, doubtless I and my audience are ignoring most of all the glasses there are in the whole wide world throughout all of time" (p. 553). Even though the sentence restricts the domain of quantification to glasses only, a statement about all the glasses in the world would be nonsensical; hence, context provides further restriction, and the quantifier is understood to range over the intersection of the set of glasses with this set, resulting, presumably, in the set of glasses held by the speaker and his friends on that occasion.

Stanley and Szábo (2000) discuss this issue at length. They write:

> What is the problem of quantifier domain restriction? Consider the sentence:
>
> (267) (Stanley and Szábo's example #1) Every bottle is empty.
>
> Suppose someone utters [(267)] in a conversation. It is unlikely that what she intends to convey is that every bottle in the universe is empty; she most likely intends to convey that every one of a restricted class of bottles (say, the bottles in the room where she is, the bottles purchased recently, etc.) is empty.
>
> p. 219

Once again, it is clear that a statement about every bottle in the universe would make little sense, and the quantifier needs to be further restricted by the context. Stanley and Szábo's focus is on the question of where the restriction provided by the context is introduced: whether it is part of syntactic structure (which is elided), a variable at the level of logical form (their preferred solution), or introduced only after pragmatic processing. Regardless of the level at which this restriction is introduced, it is clear that the motivation for postulating it in the first place is that, without it, the sentence would be unacceptable, as patently false.

5 The Generic Quantifier

Seen in this light, the generic quantifier joins syntactic traces and contextual restrictions as another empty category. Generics and habituals trigger the introduction of this empty category: **gen**.

Chierchia (1995) identifies many similarities between the generic quantifier and (overt) adverbs of quantification. Following him, and many others, I assume that the generic quantifier is a kind of adverb of quantification. Then, we would expect that, just like other adverbs of quantification, it has temporal as well as atemporal interpretations. When **gen** is used atemporally, we get a generic; and when it is used temporally, the result is a habitual. Thus, I assume that generics and habituals are similar in that they share the same quantifier (Cohen 1996); the differences between them will be discussed anon.[18]

Thus, (268a) turns into (268b), and the representation of (269a) is (269b), where **gen** is the generic quantifier.

(268) a. Birds fly.
b. Birds **gen** fly.

(269) a. John smokes.
b. John **gen** smokes.

Of course, there is a clear difference between **gen** and overt quantificational adverbs: it is phonologically null. But is this difference significant? It is often assumed in the literature that the answer is no: that **gen** is just like an overt quantificational adverb, and that the fact that it is phonologically null is of no significance.[19] However, I argue that this view is deeply problematic: the generic quantifier is not the same as overt quantifiers, and the fact that it is phonologically null has quite significant implications.

One such implication of this view is its cross-linguistic predictions. If **gen** is just another quantifier that happens to be phonologically null, we would expect to find languages where it is not null, but overt. However, there is apparently no language that has an overt element that expresses genericity (Dahl 1995).

A related problem is that if **gen** is just a phonologically null version of an overt quantifier, we would expect languages without quantifiers to lack gener-

18 But see Boneh and Doron (2008), who claim that there are, in fact, two habitual quantifiers, only one of which appears to correspond to **gen**.

19 Though we have seen some exceptions to this view (Leslie 2007b; Sterken 2015a; Nickel 2016).

ics as well. But Everett (2005) argues that Pirahã is a language without phonologically overt quantifiers, which, nonetheless, contains generics:

(270) *kaoáíbogi hi sabí 'ágahá.*
evil spirit he mean is (permanent)
'Evil spirits are mean'

If Everett's claim is granted, **gen** cannot be a mere phonologically null version of an overt quantifier.

Even if we only look at English, it turns out that there are empirical differences between overt adverbial quantifiers and generics. One difference we have already seen in chapter 3, section 4.3 above—overt quantifiers are always good with indefinite singulars, but generics are often bad (Cohen 2001b):

(271) a. *A madrigal is popular.
b. A madrigal is usually popular.

We have also seen in section 1.1 that generics can give rise to existential readings:

(272) A: I just saw a documentary about the platypus. What a funny mammal—it lays eggs!
B: Don't be silly. Mammals [don't]$_F$ lay eggs.

However, such a reading is not possible with adverbs of quantification:

(273) Mammals [don't]$_F$ usually lay eggs.

While (272) denies that *any* mammal lays eggs, and is therefore false, (273) allows that there are some mammals that lay eggs, hence it's true (Cohen 2001a).[20]

It turns out that it is actually beneficial to realize that **gen** is not really "there", and to pay close attention to the mechanisms by which it is introduced. In fact, the differences in the scopal behavior between generics and habituals are due to the fact they they use different reinterpretation mechanisms. Specifically,

20 Incidentally, these differences may account for Sterken's (2015b) observation that generics vary their interpretations more readily across different contexts, but I will not discuss this issue firther here.

habituals involve type-shifting, whereas generics involve predicate transfer. Let us consider each one of these constructions in turn.

6 Generics

6.1 *Analysis*

Recall that some generics are fine without the generic quantifier: these are cases of direct kind predication. Thus, for example, it is generally assumed that (274a) expresses no quantification, but rather the predication of the property of being an endangered species of the kind $^{\cap}$**pandas**. Hence, its logical form ought to be something like (274b).

(274) a. Pandas are endangered.
b. **endangered**($^{\cap}$**pandas**)

In contrast, other generics cannot receive this interpretation. For example, (275) does not express predication of the property of liking bamboo shoots of the kind $^{\cap}$**pandas**.

(275) Pandas like bamboo shoots.

At an intuitive level, I think this is quite clear: kinds are not the sort of things that can eat; only individual animals can eat.

But there is a stronger argument than just intuition. Consider, for example, (276a), from Carlson (1977) (his example #35, p. 165). Carlson points out that if (276a) were interpreted as predicating the property *like oneself* directly of the kind $^{\cap}$**cats**, it would have (276b) as its logical form.

(276) a. Cats like themselves.
b. **like**($^{\cap}$**cats**, $^{\cap}$**cats**)

But this logical form says that cats like cats, and we would lose the more plausible reading where, in general, a cat likes itself, but not necessarily other cats.

Consequently, in cases where direct kind predication will not do, Carlson proposes that a phonologically null generic operator is necessary. I accept this view, though for me, unlike Carlson, and like most scholars, the generic operator is a quantifier (see Cohen 1996 and chapter 3, section 2.3 for the arguments for this move). Sterken (2016) makes this argument (which she calls *the binder* *argument*, after Stanley and Szábo's 2000 binding argument) explicitly. She considers another one of Carlson's examples (his #25, p. 163):

(277) Goldfish like everyone who likes them.

Sterken points out that the object of *likes* is a variable, which must be bound by a quantifier. This quantifier is not *everyone*, which binds the subject of the verb, hence it must be the generic operator, which consequently must be a quantifier.

Thus, the logical form of (275) must include the generic quantifier; it cannot be something like (278).

(278) **like-bamboo**($^{\cap}$**pandas**)

Of course, these considerations do not preclude the proposal, made in this book, that (278) is the *initial* logical form of (275); and then the generic quantifier is introduced to produce its *final* logical form.

The question, then, is this: why must the generic quantifier be introduced? What is wrong with (278)? One possible answer is that (278) contains a type mismatch: **like-bamboo** is a predicate that applies to objects, i.e. individual pandas, and not to the kind $^{\cap}$**pandas**. This is, essentially, Carlson's (1977) view; he also draws a corresponding distinction between object-level and kind-level predicates.[21] If, indeed, type mismatch is the cause, then the operation introducing the generic quantifier ought to be type-shifting.

There is, however, little evidence for a type difference between objects and kinds. Granted, there are predicates that only apply to kinds, such as *be extinct*, *be invented*, etc. However, this shows nothing more than that these predicates have certain selectional restrictions. Almost every predicate carries selectional restrictions, some of which are extremely specific. McCawley (1976) points out that

> for example, the verb *diagonalize* requires as its object a noun phrase denoting a matrix (in the mathematical sense), the adjective *benign* in the sense 'noncancerous' requires a subject denoting a tumor, and the verb *devein* as used in cookery requires an object denoting a shrimp or a prawn.
>
> p. 67

21 More precisely, Carlson treats objects and kinds as two subtypes of individuals.

Of course, we wouldn't want to say on the basis of these selectional restrictions that matrices, tumors, and shrimp are distinct logical types!

There are, in fact, many predicates that apply to both kinds and objects:

(279) a. $\left\{\begin{array}{l}\text{Moby Dick}\\ \text{The whale}\end{array}\right\}$ is a mammal.

b. There is a a story in the newspaper about $\left\{\begin{array}{l}\text{Mary}\\ \text{tigers}\end{array}\right\}$.

c. John loves $\left\{\begin{array}{l}\text{Mary}\\ \text{tigers}\end{array}\right\}$.

d. $\left\{\begin{array}{l}\text{James Bond}\\ \text{The lion}\end{array}\right\}$ will not be forgotten.

e. $\left\{\begin{array}{l}\text{Mary}\\ \text{The lion}\end{array}\right\}$ lives in Africa.

If we maintained that kinds differ from objects in their type, we would be forced to conclude that all these predicates are ambiguous.

Hence, and for reasons of parsimony, a more attractive alternative is that both kinds and objects are of the same logical type, namely individuals; correspondingly, there is no distinction between predicates that apply to objects and predicates that apply to kinds (Cohen 1996). In the grammar, any predicate may apply to any individual. However, some of these applications will be rejected[22] due to violations of certain selectional restrictions.

Of course, selectional restrictions (or some similar devices) are needed on independent grounds, e.g. to account for the unacceptability of sentences like the following:

(280) #The rock is sad.

Sentence (281), just like (280), will be generated by the grammar, but both will be rejected because they are semantically anomalous: the property of being sad requires its argument to be animate, which a rock is not; and the property of being an endangered species requires its argument to be a kind, which an individual panda is not:

(281) #Panny, my panda, is an endangered species.

22 Or, perhaps, considered necessarily false.

In other words, the fact that kinds induce special selectional restrictions does not entail that they belong to a special logical type; world knowledge, rather than the grammar, would determine which individuals are considered to be kinds and which are not.

Let us consider (275) again. Because of the topic constraint, the BP *pandas* is first type-shifted into the kind $^{\cap}$**pandas**, resulting in the logical form (278). However, as it stands, this logical form is bad; not because of semantic mismatch, but because it violates selectional restrictions—it doesn't make pragmatic sense. We need to "save" the sentence, and the way to do it is to derive from (278), which expresses predication of a kind, a logical form that expresses quantification over instances of the kind, using the $^{\cup}$ operator. Thus, (282) is the representation of generic quantification as a standard tripartite structure.[23]

(282) $\mathbf{gen}_x[^{\cup\cap}\mathbf{pandas}(x)][\mathbf{like\text{-}bamboo\text{-}shoots}(x)]$

In general, if predication of property *P* over kind *k* does not make pragmatic sense, we derive instead:

(283) $\mathbf{gen}_x[^{\cup}k(x)][P(x)]$.

What is the process by which **gen** is introduced? If it is derived because of type mismatch, it ought to be type-shifting; but if, as I argue here, there is no type mismatch, this process ought to be predicate transfer. Which is it?

One piece of evidence that the process in question is indeed predicate transfer is the fact that introducing the generic quantifier is sometimes optional. For example, (284) is ambiguous.

(284) Animals are listed in the encyclopedia.

Under one reading, the kind $^{\cap}$**animals** is listed in the encyclopedia. But there is another reading, where, in general, if *x* is a subkind of animal, then *x* is listed in the encyclopedia. Note that both readings involve predicating a property—being listed in the encyclopedia—of kinds: either the kind $^{\cap}$**animals**, or its subkinds. However, only one of the readings involves the generic quantifier. Hence, the introduction of this quantifier is optional, rather than mandatory. This is a strong indication that the relevant phenomenon is not type-shifting, which is mandatory, since without it the sentence would be ill-formed. In contrast, as we have seen in chapter 2, section 3, predicate transfer is, in fact, optional.[24]

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23 Alternatively, we could represent this logical form using minimal situations.

24 Sterken (2016) acknowledges that if the derivation of the generic quantifier is optional,

Indeed, we can apply Nunberg's (1995) test. Recall example (49b), repeated below:

(285) I am parked out back and would like to make a phone call.

Since the argument of both conjoined predicates is the same, and since it is clearly the speaker (not the speaker's car) for the second conjunct, it follows that the argument of the predicate in the first conjunct is also the speaker. Therefore, Nunberg takes the acceptability of (285) as evidence that the predicate, but not the argument, changes meaning.

We can apply the same test to generics:

(286) Pandas like very restricted foods and are therefore an endangered species.

The fact that (286) is fine indicates that only the predicate changes meaning, not the argument: the subject of the sentence still denotes a kind, and is acceptable as an argument of a kind predicate, even though it is used in a characterizing, quantificational generic.[25]

The picture that emerges is as follows. A generic sentence is initially interpreted as predication, without the generic quantifier (cf. Cohen 2001b). If this interpretation makes sense, as in (274), nothing else needs to be done. However, if it results in a pragmatic anomaly, the generic quantifier is introduced by way of predicate transfer.

6.2 *Formalization*

In order to formalize this operation, let us generalize Dölling's (1995) definition from chapter 2, section 3.1, repeated below:

(287) $\mathscr{T}(P) =_{def} \lambda x \exists y (R(x, y) \wedge P(y))$.

this constitutes an argument in favor of the proposal that it is derived by predicate transfer. Curiously, however, Sterken uses this point as an argument *against* the derivation of **gen** by predicate transfer: the reason is that she has not been able to find cases that demonstrate the optionality of the generic quantifier. But surely (284) is precisely such an example.

25 The proposal that chacracterizing generics are generated by predicate transfer may remind some readers of Liebesman's (2011) claim that kinds can inherit properties from their instances. However, as Liebesman and Magidor (2017) point out, Liebesman's process differs from predicate transfer. More importantly, unlike the proposal made here, Liebesman denies that the result of this process in an occurrence of a quantifier in the logical form of the generic.

Recall that $\mathcal{T}$ is a predicate transfer operation, which introduces an existential quantifier, and R is a free variable over relations, whose value is left for pragmatics.

We will now define a similar operator, $\mathcal{T}_{\mathcal{G}}$, which generates the generic quantifier, and, instead of Dölling's pragmatically determined relation $R(x, y)$, the specific relation ${}^{\cup}x(y)$, meaning that y is an instance (or a plurality of instances) of the kind x:

Definition 8
$\mathcal{T}_{\mathcal{G}}(P) =_{def} \lambda x.\mathbf{gen}_y[{}^{\cup}x(y)][P(y)]$.

For example, the logical form of (288a) is (288b), to which $\mathcal{T}_{\mathcal{G}}$ applies, as in (288c), resulting in (288d).

(288) a. Pandas like bamboo shoots.
b. $\mathbf{like\text{-}bamboo}({}^{\cap}\mathbf{pandas})$
c. $(\mathcal{T}_{\mathcal{G}}(\mathbf{like\text{-}bamboo}))({}^{\cap}\mathbf{pandas})$
d. $\mathbf{gen}_x[{}^{\cup\cap}\mathbf{pandas}(x)][\mathbf{like\text{-}bamboo\text{-}shoots}(x)]$

The generation of the generic quantifier via predicate transfer can explain the scopal behavior of generics. Consider again one of Schubert and Pelletier's (1987) examples:

(289) Storks have a favorite nesting area.

Initially, this sentence is interpreted as predication, so its logical form is:

(290) $\exists x(\mathbf{favorite\text{-}nesting\text{-}area}(x) \wedge \mathbf{have}({}^{\cap}\mathbf{storks}, x))$.

However, this sentences doesn't make sense, since only individual storks, and not the kind, can have a nesting area. Hence, predicate transfer applies.

Recall that, in chapter 2, section 3 above, we have established that predicate transfer can apply at any stage of the derivation, generating scope ambiguities. One possibility is to apply it locally, as in (291a). This will result in (291b), where the generic taks narrow scope.

(291) a. $\exists x(\mathbf{favorite\text{-}nesting\text{-}area}(x) \wedge (\mathcal{T}_{\mathcal{G}}(\lambda y.\mathbf{have}(y, x)))({}^{\cap}\mathbf{storks}))$
b. $\exists x(\mathbf{favorite\text{-}nesting\text{-}area}(x) \wedge \mathbf{gen}_y[{}^{\cup\cap}\mathbf{storks}(y)]$
$[\mathbf{have}(y, x)])$